



PCA-based magnification method for revealing small signals in video

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Abstract

Video magnification techniques are useful for visualizing small changes in videos. Current methods are mainly applied for two aspects: motion amplification and color amplification. For instance, Eulerian video magnification (EVM) has shown impressive results in the context of color of human face and subtle head motion caused by the influx of blood at each beat. Such visual results have possible applications in non-contact human physiological parameter measurement, such as heart rate estimation. Unfortunately, the linear EVM is sensitive to noise and frequencies of the changes should be customized, which generates a limitation of applications. This paper presents an advanced EVM for magnifying the signal amplitude in the presence of relatively high noise as well as unknown the frequencies of changes in video. Principal component analysis (PCA) is performed to decompose the frames and the component whose spatial variation best matches small changes to be magnified. The advantage of PCA-based method is that it can select the subtle signals with a denoising process like spatial filtering. Experimental results show that the PCA-based EVM can support larger amplification factors for small changes visualization as well as less noise and artifacts.

Keywords Eulerian video magnification (EVM) · Principal component analysis (PCA) · Spatiotemporal analysis · Motion magnification

1 Introduction

Small temporal variations that are hard to see with the naked eyes may be informative and useful [1–4]. Variations in skin color caused by reaction to the flow of blood in the human face [1] or human head wobble with each heart beat [5]. Tiny motions of blood vessels in human wrist imply pulse or ulnar and radial arteries [6,7]. Low spatial amplitude fluctuation of chest can be used for respiratory rate estimation [7]. Although these small changes are usually impossible to notice [6,8–12], they can be magnified computationally to become visible and have possible applications in remote monitoring or screening for diseases [11,13].

Current video magnification techniques can be divided into two main categories: Lagrangian perspective [5,8]

and Eulerian perspective [6,9,11,12,14]. For Lagrangian approaches [8], motions are amplified by computing and tracking the trajectory of particles which easily introduce artifacts and errors. On the other hand, Eulerian approaches, such as linear techniques [6], phase-based [9] and Riesz methods [14], and video acceleration magnification [12], do not estimate motions explicitly. Instead, they amplify motions by processing and enhancing the changes of signal that evolve in frame over time. The success of these methods following Eulerian perspective is more robust and motivates the development of video magnification to reveal unperceivable information. Again, in terms of applications, current methods are also applied for two aspects: motion amplification and color amplification. Lagrangian approaches [8] and phase-based method [9,14] can only amplify small motions in videos, while linear Eulerian video magnification [6] and its advanced methods [11,12] can amplify subtle motions as well as subtle color changes. Besides, the recent work in [15] presents a method to magnify small motions on all pixels located at the same depth, by using depth cameras and bilateral filters. It aims to deal with occlusion and large motions in video magnification. The downside of this method is, of

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course, the use of additional information such as depth information.

Our work is inspired by the linear Eulerian video magnification (EVM) [6,17] which exaggerates temporal subtle motions or color changes at fixed positions by spatial and temporal processing. This work has shown impressive results of subtle motion and color amplification and further heart rate estimation. Besides, analysis and discussion also show that the approach is fast and robust and much suited to run in real time [6,14]. Unfortunately, the linear EVM is sensitive to noise and frequencies of the changes should be customized, which generates a limitation of applications. Many interesting changes occur within the presence of relatively high noise or hard to know the signals frequencies. For instance, structures sway imperceptibly in the wind [1] and weak light. When the variations in these videos are amplified by linear EVM, they will result in large artifacts and the small changes still remain hard to see because the magnified signals are still submerged by artifacts.

To counter these issues, this paper presents an advanced EVM for magnifying the signal amplitude in the presence of noise as well as unknown the frequencies of changes in video. First of all, the input video is decomposed into different spatial frequency bands by spatial pyramid, which makes motions with low amplitude, and lower spatial frequencies (spatial pooling) could be modeled based on fluid flow analysis. The changes of interest are then isolated and extracted for next amplification step. For this, principal component analysis (PCA) [5,16] is used on video sequences where each frame is treated as a sample data and its dimensionality is the number of pixels in the frame. Then, we select the components whose spatial variations can best match the changes of interest. The advantages of PCA-based method can select the subtle signals with a denoising process like spatial filtering. Finally, the small signals of interest are exaggerated via multiplying by a factor; then, the magnified ones and the original frames are combined to reconstruct the final output. We show that the proposed method can dramatically reduce noise.

The remainder of this paper is structured as follows. In Sect. 2, the PCA-based Eulerian video magnification is described. First, the processing framework of the PCA-based EVM is illustrated. We then define the problem and extract signals of interest by PCA decomposition. Section 3 demonstrates the effectiveness of the presented approach on several examples, followed by analysis of the experimental results and the comparison with the EVM approach in handling noise. Limitations of the presented approach are discussed in Sect. 4. Finally in Sect. 5, we draw conclusions of the paper.

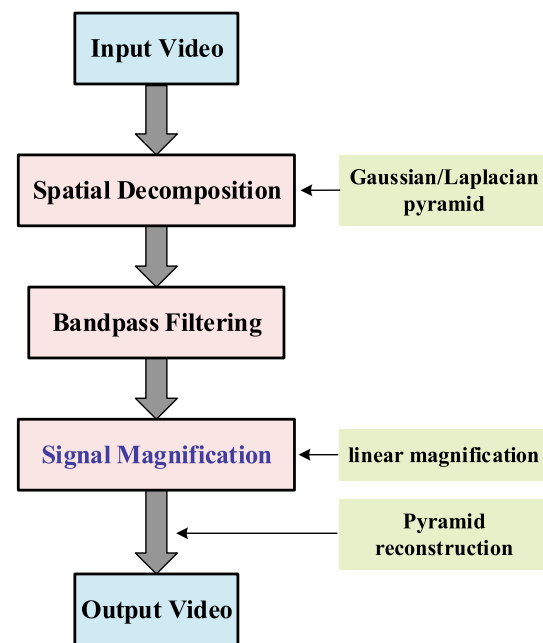


Fig. 1 Overview of linear Eulerian video magnification framework

2 PCA-based Eulerian video magnification

Linear video magnification has been presented and employed for many aspects because it is fast, robust and easily runs in real time. The framework is illustrated in Fig. 1. Eulerian method first decomposes video sequence into different spatial frequency bands by spatial pyramid, which makes motions with low amplitude, and lower spatial frequencies (spatial pooling) could be modeled based on fluid flow analysis. Then, a temporal filter is designed to perform on each spatial band to the frames and the frequency band of interest is extracted by the designed band-pass filtering processing. Furthermore, the extracted signals after band-pass filtering are multiplied by an enlargement factor or a series of ones. Finally, magnified variational signals and original frames are combined to reconstruct the final output.

The PCA-based EVM mainly deals with two problems: (i) the noise can be amplified significantly when tiny small changes are magnified with a large magnification factor. (ii) The frequencies of small changes need to be specified by the user. The processing flow of the proposed method is similar to the linear EVM. After decomposing the input into different spatial frequencies, the changes of interest are isolated and extracted for next amplification. For this, PCA is used on video sequences where each frame is treated as a sample data and its dimensionality is the number of pixels in the frame. We then select the components whose spatial variations best match the changes of interest. The signals are projected on these components, and the small changes are obtained. At last, the magnified signal is added to the original video and

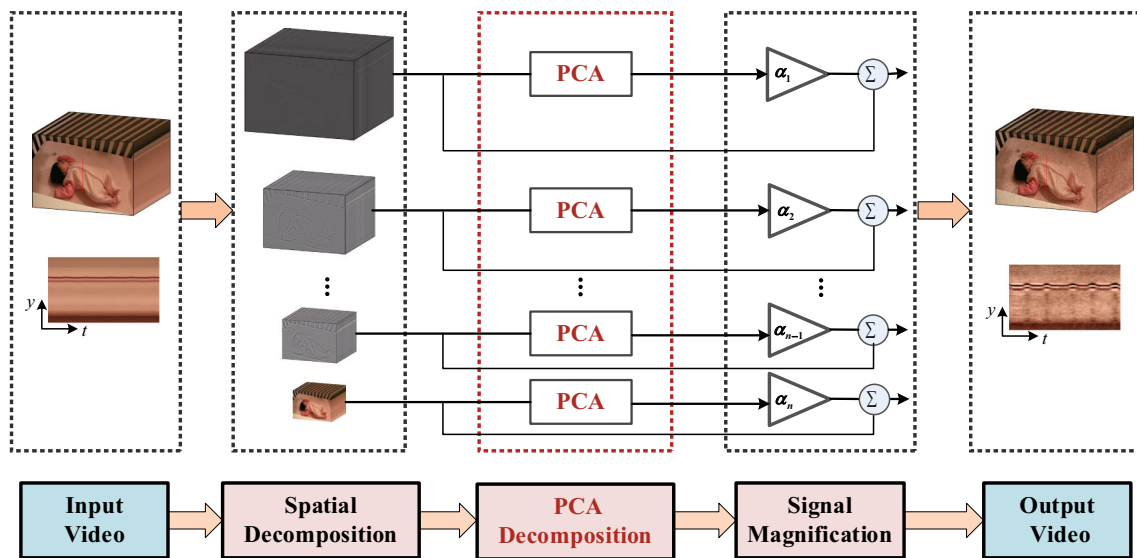


Fig. 2 Overview of the PCA-based EVM framework

the spatial pyramid is collapsed to rebuild the output video. For example, we use the system to reveal unseen motions of blood vessels in human wrist imply pulse or ulnar and radial arteries, caused by the reaction to the influx of blood at each beat. Figure 2 illustrates the overview of the PCA-based EVM.

2.1 Problem definition

The input video is assumed that small changes exist in a stationary scene. We further assume that the video scene is smooth, so that pixels within small neighborhoods exhibit similar noise levels. This assumption can be enforced in real videos by excluding parts of the scene that contain high gradients. A recorded video is modeled, including parts of small changes that contain the interference like Gaussian noise.

Let $I(x, y, t)$ denotes frame intensity at position (x, y) and time t . We model an input video frame as follows:

$$I(x, y, t) = I_0(x, y) + \psi(x, y, t) + n(x, y, t) \quad (1)$$

where $I_0(x, y)$ denotes the mean of observed image intensity, $\psi(x, y, t)$ is the signal of interest and $n(x, y, t)$ is the interference introduced during the process of video acquisition, such as Gaussian noise. These terms are assumed to be independent. Similarly, residual video frame is modeled as:

$$I_r(x, y, t) = \psi(x, y, t) + n(x, y, t) \quad (2)$$

We assume that the input video depicts a stationary scene containing signal with small changes. We further assume that the video scene is smooth in space. The level of interference is low relative to the small changes in residual terms $I_r(x, y, t)$.

2.2 PCA decomposition

The underlying source signal of interest is the small changes like color change caused by the cardiovascular pulse or kinds of motion impossible to see with the naked eye. The residual frames are a mixture of these changes as well as other interference caused by video recording. We must decompose these mixed data into subsignals to isolate the small changes of interest. To do this, we consider the multidimensional data of each residual frame as a separate data and use PCA to find a set of main dimensions which imply the useful signal. Some dimensions are selected to obtain the small changes data.

Formally, given K frames for a $M \times N$ video, we represent the K -dimensional data of residual video along with $k = M \times N$ points as $\mathbf{I}_r = [I_{r1}(k), I_{r2}(k), \dots, I_{rK}(k)]$. The mean and the covariance matrix are:

$$\bar{I}_r = \frac{1}{|K|} \sum_{i=1}^K I_{ri} \quad (3)$$

$$\Sigma_{I_r} = \frac{1}{K} \sum_{i=1}^K (\mathbf{I}_r - \bar{I}_r) (\mathbf{I}_r - \bar{I}_r)^T \quad (4)$$

PCA finds the principal axes of variation of small meaningful changes as the eigenvectors of the covariance matrix:

$$\Sigma_{I_r} \Phi_{I_r} = \Phi_{I_r} \Lambda_{I_r} \quad (5)$$

where Λ_{I_r} denotes a diagonal matrix of the eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_p$ corresponding to the eigenvectors in the columns of $\Phi_{I_r}, \Phi_1, \Phi_2, \dots, \Phi_p$. Each eigenvector represents the K -dimensional direction and magnitude of video frame points. The eigenvectors differ for each position. We obtain

Table 1 Parameters for representative videos

Video	Magnification factor	Cutoff frequencies	FPS
Baby	50	$\omega_l = 0.4, \omega_h = 3$	30
Wrist1	30	$\omega_l = 0.4, \omega_h = 3$	30
Wrist2	16	$\omega_l = 0.4, \omega_h = 4$	30
Camera	30	$\omega_l = 45, \omega_h = 100$	300
Engine	50	$\omega_l = 50, \omega_h = 650$	50

each L-dimensional (L is set by user) signal $S_{iI_r}(k)$ by projecting the data onto ϕ_i :

$$S_{iI_r}(k) = \begin{pmatrix} I_{r1}(k) \\ I_{r2}(k) \\ \vdots \\ I_{rK}(k) \end{pmatrix} \cdot \phi_i \quad (6)$$

Compared with the standard EVM, the main contribution of presented PCA-based EVM is that the interested small signals are extracted with a denoising process like spatial filtering at the same time. Thus, relatively clean signals can be obtained for magnifying by multiplication coefficient in PCA domain.

3 Experimental results

The proposed method is demonstrated on real videos as well as synthetic ones with noise added by user. Experimental results were generated using a non-optimized MATLAB code on a desktop computer with Inter(R) Core(TM) i7-3770 CPU and 16 GB RAM. It takes a few minutes to process each video. Experimental parameters, including the magnification factor α , spatial frequency cutoff λ_c and low-pass filters with cutoff frequencies ω_l and ω_h , are referred to [6] for involved videos as given in Table 1. For all videos, we process the video frames in YIQ color space. We provide representative videos and results for depicting our magnification method.

3.1 Comparison on real videos

Figure 3 displays that a newborn baby chest heaving [6] is magnified with magnification factor $\alpha = 50$. Notice that the EVM technique can magnify the respiratory movement, but aggravate the influence of both the noise and artifacts. In comparison, the new PCA-based EVM method supports larger magnification factor with fewer artifacts and less noise. Figure 4 reveals tiny motions of blood vessels in an adult right wrist with magnification factor $\alpha = 30$. The source video (a) is taken with a standard point-and-shoot camera, which is motion-magnified by the linear method (EVM) and the

new PCA-based EVM as shown in (b) and (c), respectively. The magnified pulsing arteries become more visible when observing spatiotemporal X-T slices from the input (b) and (c) (below). The PCA-based EVM yields good results with less noise when amplification factor is 30. The results of all representative frames demonstrate that PCA-based EVM can extract and magnify small variations in videos and obtain better performance.

3.2 Comparison on synthetic video

In some videos, the variation of signal of interest is usually contaminated by noise and interference. In such videos, direct enhancement of the pixel values is hard to reveal the desired signal in the linear EVM. We thus consider to generate a synthetic video added some Gaussian white noise with respect to space to verify the algorithms. The synthetic video contains blobs oscillating and the variance of added zero-mean Gaussian is 0.002 with all numerical parameters normalized. Figure 5 shows the results amplified by two different factors $\alpha = 10$ and 20. A frame in a sequence in the presence of noise is displayed in (a); the motion is amplified by linear EVM in both (b) and (d); accordingly, (c) and (e) show the output using PCA-based method. We note that PCA-based EVM approach can get better results in noise handling.

3.3 Denoising analysis

Our interested signal commonly suffers from a degree of contamination when a magnification is implemented, especially with a large magnification factor. Figures 6 and 7 show our results for baby (Fig. 3) and wrist (Fig. 4), where different positions are selected in videos. In Fig. 6, the line graphs illustrate intensity traces over time for the pixel marked on the input frame in (a) at position A, B, C, D, where the black line shows the trace from the original sequences, and the blue and red one show the trace using the linear and PCA-based EVM, respectively. The respiration signal is hardly visible in blue line as the noise level is higher than the power of the signal, while in red line the respiration signal is easily visible and the periodic peaks can be seen clearly in the trace. Specifically, in (e), PCA-based method hardly amplifies the noise, while the linear magnification method amplifies the noise linearly in a factor. Similarly, in Fig. 7, different colored lines display the intensity trace over time from original sequences, the linear and PCA-based magnification result. From the result of position A labeled on the input sequence, the pulse signal can be observed clearly in red line, while it is hard to see in blue line. In (c) and (d), the intensity traces indicate that the linear EVM amplifies the noise in background, while the noise is hardly magnified using PCA-based method.

Figure 8 shows plots of noise magnified for the linear and PCA-based method by two different magnification factors.

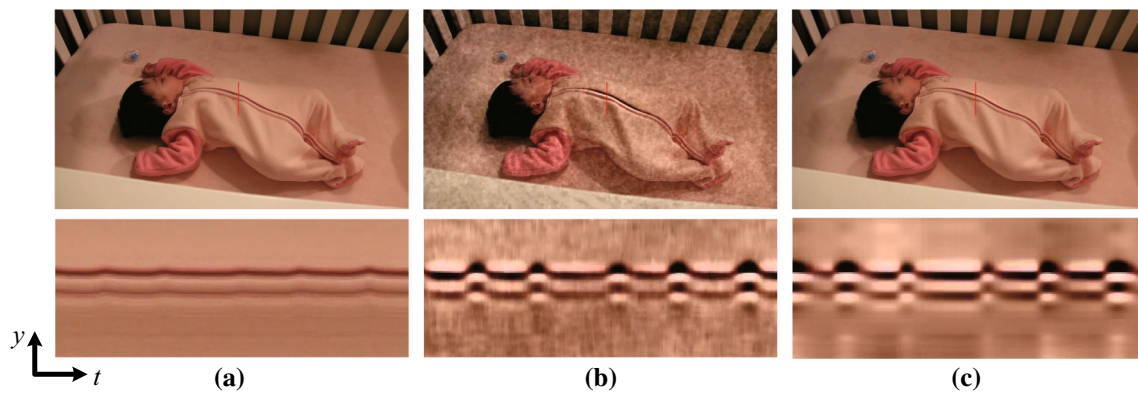


Fig. 3 Comparison of the motion magnification results on baby sequences. **a** Top: a original frame from the input video (baby) displayed on the left. Bottom: a spatiotemporal Y-T slice along red line marked on the input frame is shown. **(b, c)** the EVM [6] and the PCA-

based EVM (this paper) motion processing results, respectively, and the corresponding spatiotemporal slices as shown in the middle and right. **a** Source, **b** EVM, **c** PCA-based EVM (color figure online)

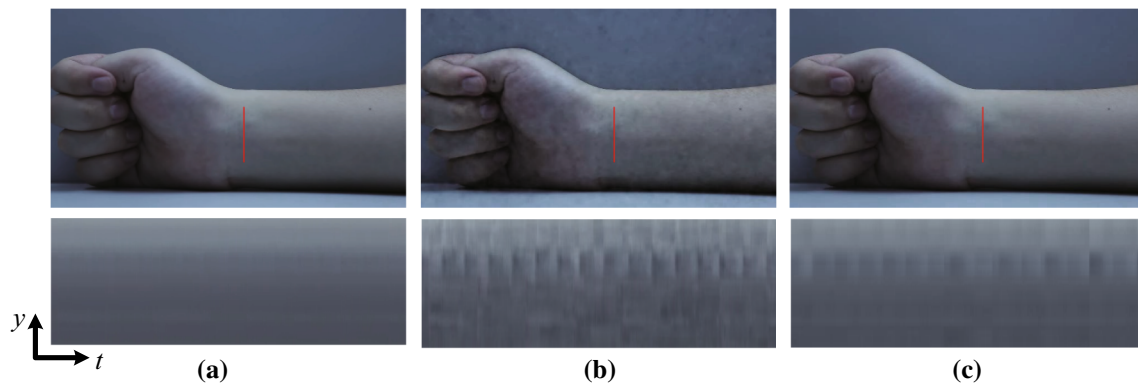


Fig. 4 Motion magnification results of subtle motions of blood vessels. **a** Source, **b** EVM, **c** PCA-based EVM

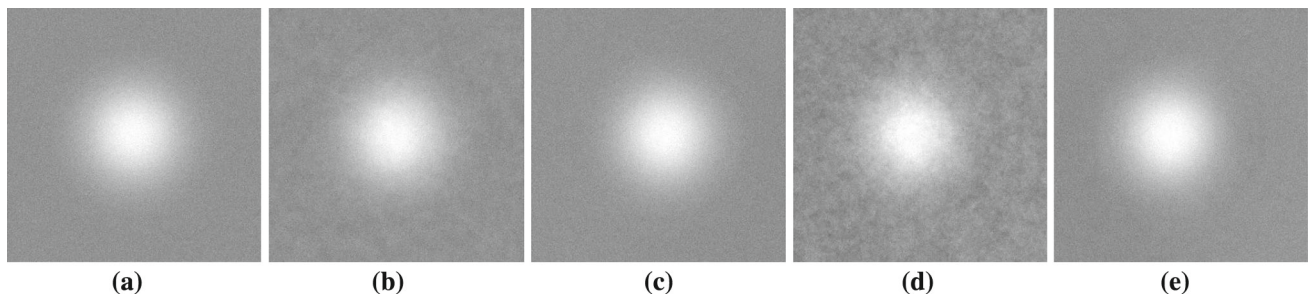


Fig. 5 Comparison of our result on the synthetic sequences suffered from Gaussian white noise with the result of the EVM using two different factors $\alpha = 10$ and 20. **a** Input noisy frame, **b** EVM, $\alpha = 10$, **c** PCA-based EVM, $\alpha = 10$, **d** EVM, $\alpha = 20$, **e** PCA-based EVM, $\alpha = 20$

A stationary range marked in (a), which should contain no change signal, is selected to exhibit the intensity of error. Our results in (b) and (c) indicate that the level of noise is similar to the original noisy video (as can be seen in black line and the red one), while noise observed in results using the linear method is much larger (as can be seen in blue line). In other words, the results demonstrate that noise can not be amplified as the amplification factor is increased in PCA-based method, which is better than the linear EVM.

4 Discussion and limitations

The disadvantage of presented PCA-based EVM as well as the standard EVM assumes that there is a locally single small signal. Thus, many interesting small deformations occurred within or because of larger motion or kinds of changes are magnified causing motion blur. For this issue, the dynamic video motion magnification (DVMAG) [11] and video acceleration magnification [12] have dealt with and

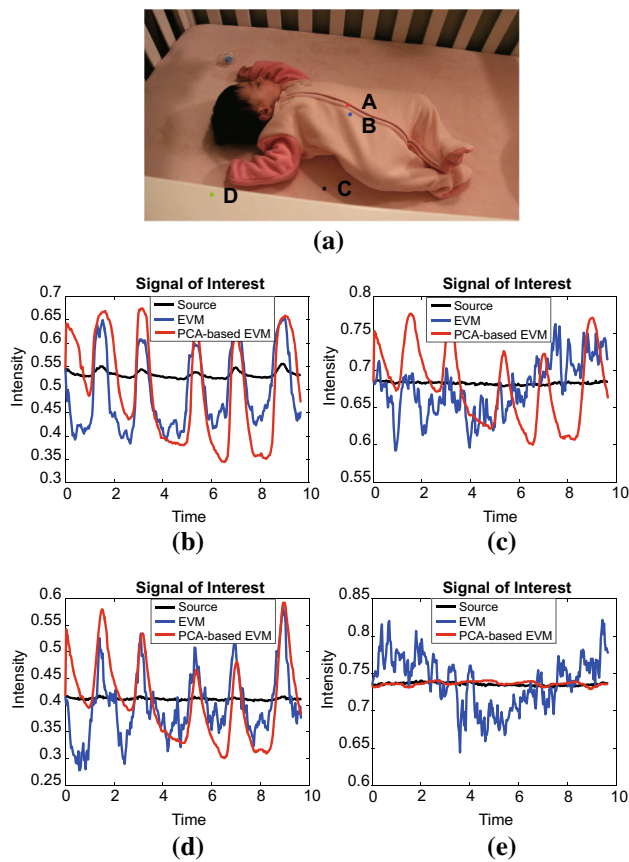


Fig. 6 Comparison between the linear and PCA-based EVM for baby chest heaving in handling noise. **a** A frame from the baby video. In the bottom (from Figure **b** to **e**) are intensity traces over time for the pixel at different positions marked on the input frame in **(a)**. **a** One frame of input video, **b** signal at position A, **c** signal at position B, **d** signal at position C, **e** signal at position D

acquired some results using different approaches. The limitation of the DVMAG is that regions of interest should be selected by user and the video acceleration magnification assumes that the large motion should be linear. There is still some room for improvement. Other limitations of our method are that there is a background dragging effect around object boundaries. Therefore, nonlinear amplification parameters should be designed for the high gradient in image sequences. In addition, amplitude-selective filtering is employed for the periodic small changes in the presence of large motions with similar frequencies in future work.

5 Conclusion

This paper presents a video magnification method based on PCA to reveal small changes of interest in videos. It can magnify the signal amplitude in the presence of relatively high noise as well as unknown the frequencies of the changes. The main contribution of presented PCA-based EVM, compared

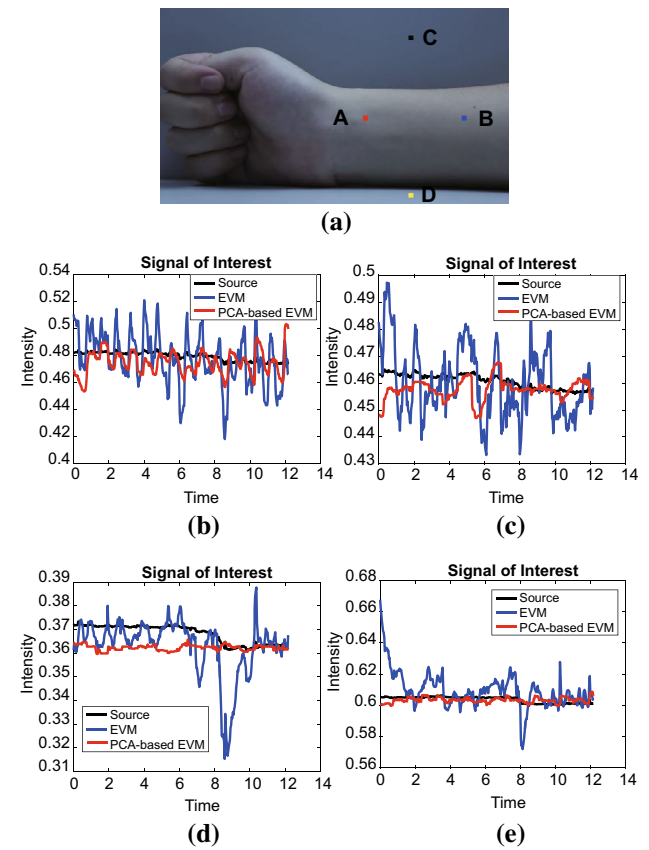


Fig. 7 Comparison between the linear and PCA-based EVM for tiny motions of blood vessels in handling noise. **a** One frame of input video, **b** signal at position A, **c** signal at position B, **d** signal at position C, **e** signal at position D

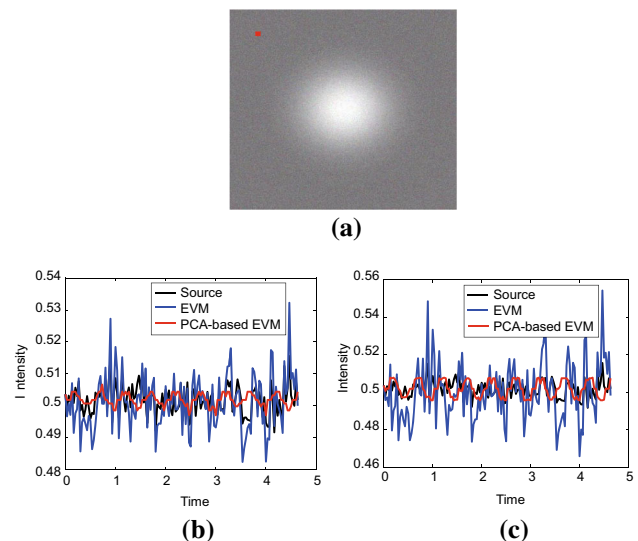


Fig. 8 Comparison between the linear EVM and PCA-based video magnification for synthetic sequences in handling noise. **a** Input noisy frame, **b** magnification by $\alpha = 10$, **c** magnification by $\alpha = 20$

with the standard EVM, is extracting interested small signals by mathematical tools with a denoising process like spatial

filtering. Thus, relatively clean signals can be obtained for magnifying by multiplication coefficient in PCA domain. Experimental results show that the proposed method can support larger amplification factors for small changes visualization with less noise.

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